WORKED SOLUTION

STEP II 2003, Question 5

Pure • Vector Geometry of Lines

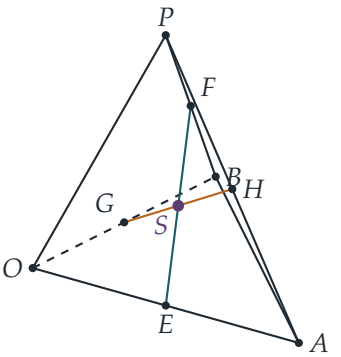
STEP II 2003, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q5

The key idea

Write down the four midpoints, put each line in the form "point + parameter × direction", and equate. Two lines in three dimensions usually miss each other, so the work lies in showing that the three component equations in two unknown parameters are consistent. The **k**-components settle things quickly. Geometrically, O, A, B, P form a tetrahedron, and EF and GH each join the midpoints of a pair of opposite edges. That suggests they meet at the "average" of the four vertices.



Solution

The lines EF and GH meet

Write $\mathbf{p} = l\mathbf{i} + m\mathbf{j} + n\mathbf{k}$. The midpoints have position vectors

$$\mathbf{e} = \tfrac{1}{2}a\mathbf{i}, \qquad \mathbf{f} = \tfrac{1}{2}(l\mathbf{i} + (b+m)\mathbf{j} + n\mathbf{k}),$$
$$\mathbf{g} = \tfrac{1}{2}b\mathbf{j}, \qquad \mathbf{h} = \tfrac{1}{2}((a+l)\mathbf{i} + m\mathbf{j} + n\mathbf{k}).$$

A general point of EF is $\mathbf{e} + s(\mathbf{f} - \mathbf{e})$, and a general point of GH is $\mathbf{g} + t(\mathbf{h} - \mathbf{g})$. Doubling both to clear the halves,

$$2(\mathbf{e} + s(\mathbf{f} - \mathbf{e})) = (a + s(l - a))\mathbf{i} + s(b + m)\mathbf{j} + sn\,\mathbf{k},$$
$$2(\mathbf{g} + t(\mathbf{h} - \mathbf{g})) = t(a + l)\mathbf{i} + (b + t(m - b))\mathbf{j} + tn\,\mathbf{k}.$$

The lines meet if and only if there are values of s and t that make these equal, so we need all three of

$$\begin{aligned} \mathbf{i}: &\quad a + s(l - a) = t(a + l), \\ \mathbf{j}: &\quad s(b + m) = b + t(m - b), \\ \mathbf{k}: &\quad sn = tn. \end{aligned}$$

Ask yourself Three equations in two unknowns: why should they have a solution at all?

In general they don't, and that's why two lines in space usually miss each other. Here we solve two of the equations and check that the third holds too. Since $n \neq 0$, the $\mathbf{k}$-equation gives $s = t$. The $\mathbf{i}$-equation then becomes $a + sl - sa = sa + sl$, so $a = 2sa$, and since $a \neq 0$ we get $s = \frac{1}{2}$. Finally, with $s = t = \frac{1}{2}$ the $\mathbf{j}$-equation reads

$$\tfrac{1}{2}(b + m) = b + \tfrac{1}{2}(m - b) = \tfrac{1}{2}(b + m),$$

which is true. So all three equations hold when $s = t = \frac{1}{2}$, and the lines EF and GH intersect, as required.

The point of intersection is the midpoint of EF (and also of GH):

$$\mathbf{s} = \tfrac{1}{4}((a + l)\mathbf{i} + (b + m)\mathbf{j} + n\mathbf{k}) = \tfrac{1}{4}(\mathbf{0} + a\mathbf{i} + b\mathbf{j} + \mathbf{p}),$$

the average of the position vectors of O, A, B and P.

The point T

Since S is the midpoint of DT, we have $\mathbf{s} = \frac{1}{2}(\mathbf{d} + \mathbf{t})$, so $\mathbf{t} = 2\mathbf{s} - \mathbf{d}$:

$$\mathbf{t} = \tfrac{1}{2}(a + l)\,\mathbf{i} + \tfrac{1}{2}(b + m)\,\mathbf{j} + (\tfrac{1}{2}n - d)\mathbf{k}.$$

The plane OAB contains O and the vectors $a\mathbf{i}$ and $b\mathbf{j}$. Since a and b are non-zero, these point along the x- and y-axes, so the plane OAB is the plane $z = 0$. So T lies in it exactly when its $\mathbf{k}$-component vanishes, that is, $\frac{1}{2}n - d = 0$. Hence

$$d = \tfrac{1}{2}n.$$

This is non-zero because $n \neq 0$, so it is consistent with the condition $d \neq 0$.

Checks

- Take $a = 2$, $b = 4$, $\mathbf{p} = (2,0,4)$. Then $E = (1,0,0)$, $F = (1,2,2)$, $G = (0,2,0)$, $H = (2,0,2)$. The midpoints of EF and GH are both $(1,1,1)$, which is $\frac{1}{4}\big((0,0,0) + (2,0,0) + (0,4,0) + (2,0,4)\big)$.
- With $d = 2 = \frac{1}{2}n$, $T = 2(1,1,1) - (0,0,2) = (2,2,0)$, which lies in $z = 0$.
- A computer check confirmed, in exact rational arithmetic for 20 000 random choices of a, b, l, m, n, d, that the lines meet at $s = t = \frac{1}{2}$ at the point above, and that T lies in the plane OAB (scalar triple product zero) exactly when $d = \frac{1}{2}n$.

Common slips

- Equating the two lines with the *same* parameter from the start. That assumes the answer $s = t$, which here comes from the **k**-equation.
- Solving two of the component equations and stopping. Showing that two lines in space intersect needs the third equation checked as well.
- Forgetting to say where $a \neq 0$ and $n \neq 0$ are used: they are what let us divide to find s and t.

Beyond STEP

The three lines joining midpoints of opposite edges of any tetrahedron (the *bimedians*) all pass through its centroid, the average of the four vertices, and each is bisected there. This is the three-dimensional cousin of the medians of a triangle meeting at the centroid, and it is a first taste of thinking about points as weighted averages, the idea behind barycentric and affine coordinates. Strang's *Introduction to Linear Algebra* develops the vector and linear-equation viewpoint used here, including when a system with more equations than unknowns is consistent.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.